LITERATURE SURVEY AND REQUIREMENT ANALYSIS REPORT

REVA Internship Opportunities (RIO) | School of Computer Science and Engineering, REVA University, Bengaluru

Particulars	Details
Task	Task 2 (SL 2), Project Track: Literature Survey and Requirement Analysis of the Module
Name of Student	Kartik Jarali
University Seat Number (USN)	R23EF107
Project Title	NextTo: A Hyperlocal Deals Marketplace for Local Businesses and Consumers
Track	Other
Centre	Center for Cyber Security and Digital Trust
Faculty Guide	Dr Chaithra M H
Date of Submission	9 October 2026

1. Introduction

1.1 Purpose. This report records the literature survey carried out for the NextTo project and the functional and non-functional requirements derived from it. Its purpose is to show what existing research offers for a hyperlocal deals marketplace, where that research falls short, and how the shortfalls shape the requirements of the system.

1.2 Background. NextTo is a marketplace in which local businesses publish discounted offers and nearby customers discover, buy and redeem them in person. The platform consists of consumer and merchant mobile applications, a consumer website and an administrator dashboard, supported by a shared Supabase and PostgreSQL backend with UPI payments.

1.3 Scope. The survey covers the whole platform, because the assigned module depends on the other modules. It is organised under the following themes: hyperlocal marketplace concept and business model; location-based deal discovery and recommendation; backend and marketplace architecture; and UPI payment security.

1.4 Method. Candidate papers were searched in the IEEE Xplore database against the project problem statement. Seventeen candidate papers were identified, grouped by theme. Ten of these were read in full and are reviewed in Section 2. For each paper, the approach, the principal findings and the limitation relative to NextTo are recorded.

2. Literature Survey

The ten papers reviewed are summarised below. For each paper, the approach, the key findings and the limitation relative to NextTo are described in turn. The number in brackets corresponds to the reference number in Section 6.

[1] Vas et al., 2023 (ICAC3N). This conceptual paper defines a hyperlocal marketplace, its stakeholders (suppliers, customers, delivery partners and an administrator) and a phased cloud architecture. It identifies the platforms such a marketplace needs, namely a website, customer and vendor apps and an administrator panel, proposes a phased rollout, and lists advertising and commission as revenue models. However, it remains purely conceptual: it gives no implementation, data model, security or payment detail, and it does not cover service or appointment-based deals or voucher redemption.

[2] Kohli et al., 2024 (ISCS). The authors migrate a monolithic e-commerce platform to Python and Flask microservices (authentication, cart, product and image) running on Kubernetes, with DevOps CI/CD. The services scale independently, user satisfaction exceeded 90%, the system recovers from pod failure, and read and write bandwidth improved under load. The work does not, however, address separate merchant, consumer and administrator roles, payment and voucher domains, or location services.

[3] Cecil et al., 2024 (ICC-ROBINS). This system combines smart-cart location tracking with decision trees, collaborative filtering, content-based filtering and FP-Growth association rules to recommend products in store. Revenue and dwell time increased after deployment, and the hybrid of filtering methods outperformed any single method. The design is built for physical retail with smart carts and tags rather than a mobile-app deals marketplace, and it has no online purchase or voucher flow.

[4] Sharma and Kaur, 2015 (ICCCA). The authors present a rule-based, context-aware recommender driven by user-defined rules such as weather, time and distance. Places are ranked by a weighted function of distance, rule priority, tag preference and public rating, and rule-based pre-filtering reduced a 100-place data set to 8 relevant places before ranking. Its weakness is that the user must write the rules, so it does not scale to large catalogues or live deal stock; it is also dated and was not designed for mobile.

[5] Satrio et al., 2022 (ICoICT). This qualitative customer-development study interviewed buyers and sellers in two Indonesian cities to derive a validated minimum viable product. It produced 36 buyer-app and 35 seller-app features, confirmed that loyalty points are a must-have feature, and showed regional differences in payment and delivery. The feature list is, however, general to marketplaces, and vouchers, QR redemption and appointments were not validated.

[6] Lin and Hsu, 2018 (IC3). The paper proposes a five-module architecture (event analysis, preference analysis, store character, store fitness and store ranking by AHP) that recommends stores according to why a user is at a location. Its contribution is to recommend by event and social influence rather than by location alone. It is conceptual only, with no data set or experiment, and it needs a social graph that NextTo does not plan to build.

[7] Tong et al., 2017 (IEEE Big Data Analysis Conf.). The authors convert next-basket recommendation into a classification problem, using feature selection, time-decay functions and geohash-based imputation of missing location. On the Alibaba mobile-commerce data, the hybrid method reached an

F1 score of 11.15%, against 9.00% for logistic regression alone. The absolute scores are low, and the geohash method gives up location precision, so it does not suit exact-radius deal search.

[8] de O. Junior et al., 2022 (ICCE). This paper presents a microservices reference architecture with authentication, register, user and configuration services, an API gateway and a message broker, secured with OAuth and JWT. The result is a modular, independently deployable design that is easier to maintain than a monolith. However, it includes no performance testing and no marketplace services such as payments or vouchers, and the authors state that it is unproven against a monolith.

[9] Chang, 2010 (3PGCIC). The paper describes Application Marketplace as a Service, a reference architecture with private-label multi-tenancy, ten functional blocks and twenty-six service interfaces. It shows that one core system can serve several branded marketplaces on a shared backend, which is similar to serving many merchants on one backend. The work is dated and predates REST and serverless practice; it also concerns downloadable software rather than physical deals and has no location logic.

[10] Lingareddy et al., 2025 (GINOTECH). The authors build a hybrid DQN (reinforcement learning) and XGBoost model for UPI fraud detection, with SMOTE balancing, KNN imputation and SHAP feature importance. The model achieved 98.7% accuracy, 97.9% precision, 98.5% recall, a 1.2% false-positive rate and a 45 ms inference time. It was, however, trained on a public Kaggle data set rather than live UPI data, and the 45 ms figure excludes network and payment gateway delay.

3. Literature Gaps Identified

3.1 Grouping of the Literature. The ten papers fall into three groups, each relating to a part of NextTo.

- **What a marketplace needs.** Papers [1], [5] and [9] describe the roles, minimum features and shared-backend model of a hyperlocal or multi-tenant marketplace. They are conceptual or dated, and none shows a modern serverless implementation.
- **How the backend is built.** Papers [2] and [8] describe microservice and gateway designs. Both were tested on general e-commerce or device settings, not on a location-aware deals marketplace.
- **How deals are discovered and paid for.** Papers [3], [4], [6] and [7] show a progression from simple distance ranking in 2015 to hybrid machine learning in 2024. All of them depend on rich history, social data or special hardware that a new platform will not have. Paper [10] is the only one on payment security, and the gap between its benchmark results and a real payment flow remains open.

3.2 Gaps and the Response of the Project. Each gap is matched below with the decision taken for NextTo.

Gap 1: Vouchers and appointments. No existing work covers vouchers, QR redemption or appointment-based deals in a hyperlocal setting (papers [1] and [5]). NextTo therefore treats voucher issue, single-use redemption and slot booking as core requirements (FR-C4, FR-C6, FR-M3 and FR-M4).

Gap 2: Dated marketplace architectures. Marketplace architectures in the literature are conceptual or dated, with no serverless implementation (papers [1] and [9]). NextTo uses a shared Supabase and PostgreSQL backend with server-side Edge Functions grouped by domain.

Gap 3: Incomplete microservice designs. The microservice designs studied omit roles, payments, vouchers and location services (papers [2] and [8]). NextTo responds with independent domain modules, token-based login and role-based access control (NFR-2 to NFR-4).

Gap 4: Recommenders that need history. The recommenders reviewed depend on user history, rules written by users or social data, none of which exist at launch (papers [3], [4], [6] and [7]). NextTo instead uses a simple, explainable ranking by distance, category, recency and urgency that works with no history (FR-D1 and NFR-6). Personalised ranking is left for later.

Gap 5: Unreliable location handling. Existing location handling either trades accuracy for privacy or assumes that location is always available (paper [7]). NextTo uses exact radius search, with a fallback to a chosen area when location is denied or unavailable (FR-D2 and NFR-7).

Gap 6: Unproven fraud detection. The fraud-detection model reviewed is trained on public data, and its reported speed excludes gateway delay (paper [10]). NextTo therefore builds payment behind a replaceable interface, so that a fraud model can be added in Phase 2 after validation (NFR-5).

Gap 7: Multi-merchant configuration. Multi-merchant branding and configuration have not been treated for deals (paper [9]). NextTo supports merchant-scoped configuration within one shared schema (NFR-8).

4. Requirement Analysis

4.1 Basis. The requirements below are derived from the problem statement and from the gaps in Section 3. Each requirement is traced to the problem statement or to the paper that motivates it. The identifiers are used again in the System Design Document. Prefixes are: C for consumer, M for merchant, A for administrator, D for discovery, P for platform.

4.2 Functional Requirements.

FR-C1 (Consumer). The consumer can register, log in and manage a profile. Traced to the problem statement and papers [1] and [5].

FR-C2 (Consumer). The consumer can search for nearby deals by category, keyword and distance. Traced to papers [3], [4] and [6].

FR-C3 (Consumer). The consumer can purchase a deal online and receive a unique QR or barcode voucher. Traced to the problem statement.

FR-C4 (Consumer, Merchant). The voucher is redeemed in person, and the system marks it as used exactly once. Traced to the problem statement.

FR-C5 (Consumer). After redemption, the consumer can rate and review the merchant and leave a tip. Traced to the problem statement.

FR-C6 (Consumer, Merchant). The consumer can book an appointment slot for a service-based deal. Traced to the problem statement.

FR-C7 (Consumer). The consumer earns and redeems reward points on each transaction. Traced to paper [5].

FR-C8 (Consumer). The consumer can use saved searches, deal alerts and referrals (Phase 2). Traced to the problem statement.

FR-M1 (Merchant). The merchant registers with business verification documents. Traced to paper [1].

FR-M2 (Merchant). The merchant can create a deal, submit it for approval and manage active offers. Traced to papers [1] and [9].

FR-M3 (Merchant). The merchant validates a customer voucher by scanning it or by manual entry. Traced to the problem statement.

FR-M4 (Merchant). The merchant can create appointment slots and manage bookings. Traced to the problem statement.

FR-M5 (Merchant). The merchant can view sales and revenue. Traced to paper [1].

FR-A1 (Administrator). The administrator approves or rejects merchants and deals. Traced to paper [1].

FR-A2 (Administrator). The administrator decides refund requests. Traced to the problem statement.

FR-A3 (Administrator). The administrator can view analytics, manage users and merchants, handle payouts and review the audit log. Traced to papers [1] and [9].

FR-D1 (System). The system ranks nearby deals using distance, category match, recency and urgency. Traced to papers [3] and [4].

FR-D2 (System). The system continues to rank deals by using a chosen area when location is denied or data is sparse. Traced to paper [7].

FR-P1 (Consumer, System). The system takes UPI payment through Razorpay or Cashfree and handles success, failure and refund. Traced to the problem statement and paper [10].

FR-P2 (System). The system sends order, voucher, booking and approval notifications. Traced to the problem statement.

FR-P3 (System). All applications use one common interface to the backend. Traced to the problem statement and papers [2] and [8].

4.3 Non-Functional Requirements.

NFR-1. Voucher validation completes in under about 200 ms, so that redemption at the counter is not delayed. Traced to paper [10].

NFR-2. Each service (authentication, deals, payments and vouchers) can be deployed and scaled independently. Traced to papers [2] and [8].

NFR-3. Services authenticate each other and users by tokens (JWT or OAuth). Traced to paper [8].

NFR-4. Role-based access control separates consumer, merchant and administrator permissions. Traced to paper [1].

NFR-5. The fraud-detection layer is replaceable, so a Phase 2 model can be added without redesigning payments. Traced to paper [10].

NFR-6. Ranking still works for new users and new merchants who have no history. Traced to papers [3] and [7].

NFR-7. The system tolerates missing or inaccurate location, such as denied permission or indoor drift. Traced to paper [7].

NFR-8. Merchant-specific configuration is supported without duplicating core services. Traced to paper [9].

NFR-9. Administrator actions such as approvals and refunds are logged and can be audited. Traced to papers [1] and [9].

NFR-10. The mobile applications and the website share one backend interface. Traced to the problem statement.

4.4 Notes on Scope.

- Deal ranking (FR-D1) is deliberately a simple score of distance, category, recency and urgency, because the machine-learning methods in papers [3] and [7] need history that a new platform lacks. Personalised ranking is future work.
- NFR-5 and NFR-7 exist because the fraud results in paper [10] come from a public data set, and the location method in paper [7] gives up precision. Neither is safe to adopt directly in a payment or discovery flow without further validation.
- NFR-8 follows the private-label pattern of paper [9], applied to many merchants instead of many branded software stores.

4.5 Requirements of the Assigned Module. The requirements that apply directly to the assigned module, Orders and Payments, are listed below. The module creates an order for a deal, takes UPI payment through Razorpay or Cashfree, confirms payment only through a verified gateway callback, and handles refunds. It runs server-side as Supabase Edge Functions over PostgreSQL and is shared by the consumer mobile app and the website.

- **FR-C3.** The module creates the order for a purchased deal and, once payment is confirmed, triggers issue of the unique QR or barcode voucher.
- **FR-P1.** The module is the core of this requirement: it starts the UPI payment through Razorpay or Cashfree and handles success, failure and refund outcomes.
- **FR-A2.** The module carries out the refund once an administrator approves the request, and records the result against the order.
- **FR-C7.** On a confirmed payment, the module records the transaction from which reward points are earned, and applies any points redeemed against the order total.
- **FR-P2.** The module raises order-confirmation, payment-failure and refund notifications.
- **FR-P3 and NFR-10.** The consumer app and the website call the same order and payment endpoints through the common backend interface.
- **NFR-2.** Payments run as a separate set of Edge Functions, so they can be deployed and scaled apart from deals and vouchers.
- **NFR-3.** Every order and payment call is authenticated by token, and a payment is marked successful only after the gateway callback signature has been verified on the server.

- **NFR-4.** Role-based access control lets a consumer see only their own orders and limits refund decisions to administrators.
- **NFR-5.** The gateway and risk checks sit behind a replaceable interface, so a fraud-detection model can be added in Phase 2 without redesigning the payment flow.
- **NFR-9.** Every payment status change and refund decision is written to the audit log.

5. Conclusion

The survey shows that existing research supplies ideas for marketplace roles, modular backends and location-based recommendation, but does not combine them for a hyperlocal deals marketplace with vouchers, appointments and UPI payments. The project therefore adopts a modular Supabase backend, a simple ranking method that works without history, and a replaceable payment-risk layer, and leaves personalised ranking and machine-learning fraud detection for Phase 2.

6. References

[1] O. Vas, M. Fernandes, R. Keny, S. Mascarenhas, and J. Naik, "Hyperlocal marketplace: transforming unorganized localized marketplace with future tech," in *Proc. 2023 5th Int. Conf. on Advances in Computing, Communication Control and Networking (ICAC3N)*, 2023, pp. 1596-1600, doi: 10.1109/ICAC3N60023.2023.10541378.

[2] D. Kohli, J. Kumar, J. Saini, S. Aggarwal, Y. Singh, and A. Garg, "Implementing microservice architecture in e-commerce with DevOps practice," in *Proc. 2024 Int. Conf. on Intelligent Systems for Cybersecurity (ISCS)*, 2024, doi: 10.1109/ISCS61804.2024.10581082.

[3] J. Cecil L, S. Juliet, J. Anitha, and S. M. Kurian, "Intelligent location-based recommendation system for enhancing shopping experience," in *Proc. 2024 Int. Conf. on Cognitive Robotics and Intelligent Systems (ICC-ROBINS)*, 2024, pp. 757-762, doi: 10.1109/ICC-ROBINS60238.2024.10533892.

[4] S. Sharma and D. Kaur, "Location based context aware recommender system through user defined rules," in *Proc. Int. Conf. on Computing, Communication and Automation (ICCCA)*, 2015, pp. 257-261.

[5] S. Satrio, E. K. Budiarjo, and M. R. Shihab, "Minimum viable product analysis in hyperlocal marketplace applications using customer development method," in *Proc. 2022 10th Int. Conf. on Information and Communication Technology (ICoICT)*, 2022, pp. 334-339, doi: 10.1109/ICOICT55009.2022.9914834.

[6] L.-F. Lin and Y.-L. Hsu, "Social event shopping recommendation mechanism for location-based services," in *Proc. 2018 1st Int. Cognitive Cities Conf. (IC3)*, 2018, pp. 43-44, doi: 10.1109/IC3.2018.00019.

[7] J. Tong, H. Ma, W. Liu, and B. Wang, "Time and location-based hybrid recommendation system," in *Proc. 2nd IEEE Int. Conf. on Big Data Analysis*, 2017, pp. 677-683.

[8] R. S. de O. Junior, R. C. A. da Silva, M. Souza Santos, D. W. Albuquerque, H. O. Almeida, and D. F. S. Santos, "An extensible and secure architecture based on microservices," in *Proc. 2022 IEEE Int. Conf. on Consumer Electronics (ICCE)*, 2022, doi: 10.1109/ICCE53296.2022.9730757.

[9] S.-F. Chang, "Application marketplace as a service: a reference architecture for application marketplace service," in *Proc. 2010 Int. Conf. on P2P, Parallel, Grid, Cloud and Internet Computing*, 2010, pp. 186-192, doi: 10.1109/3PGCIC.2010.32.

[10] N. Lingareddy, D. Indoria, S. Deepika, G. George, M. K. Priya, and R. Tamilarasi, "Enhancing digital payment security: UPI fraud detection with advanced machine learning algorithms," in *Proc. 2025 Global Conf. in Emerging Technology (GINOTECH)*, 2025, doi: 10.1109/GINOTECH63460.2025.11077038.